

1 Proving Q

To derive the risk-neutral probability q , we consider an investor who allocates wealth between the stock S and the bond B by choosing a portfolio $H = (\alpha, \beta)$. The portfolio value at $t = 1$ is:

$$V_1 = \alpha S_1 + \beta \cdot B_1 = \begin{cases} \alpha \cdot S_0 \cdot u + \beta B_1 = X_{\text{up}} \\ \alpha \cdot S_0 \cdot d + \beta B_1 = X_{\text{down}} \end{cases}$$

The investor maximises expected utility $\mathbb{E}^P[u(V_1)]$ subject to the budget constraint $\alpha S_0 + \beta B_0 = V_0$. We show that the first-order conditions uniquely pin down a probability measure q (the **risk-neutral probability**) under which all assets earn the risk-free rate r .

The investor's optimisation problem is thus:

$$\max_{\alpha, \beta} \mathbb{E}^P[u(\alpha S_1 + \beta B_1)] \quad \text{s.t.} \quad \alpha S_0 + \beta B_0 = V_0$$

where the constraint states that the cost of the portfolio today equals initial wealth V_0 . Since the constraint is an equality, we maximise using the Lagrangian:

$$\begin{aligned} \mathcal{L}(\alpha, \beta, \lambda) &= \mathbb{E}^P[u(\alpha S_1 + \beta B_1)] - \lambda(\alpha S_0 + \beta B_0 - V_0) \\ &= p \cdot u(X_{\text{up}}) + (1 - p) \cdot u(X_{\text{down}}) - \lambda(\alpha S_0 + \beta B_0 - V_0) \\ &= p \cdot u(\alpha S_0 \cdot u + \beta B_1) + (1 - p) \cdot u(\alpha S_0 \cdot d + \beta B_1) - \lambda(\alpha S_0 + \beta B_0 - V_0) \end{aligned}$$

Taking first-order conditions with respect to α and β :

$$\begin{aligned} \mathcal{L}_\alpha &= p \cdot u'(X_{\text{up}}) \cdot S_0 \cdot u + (1 - p) \cdot u'(X_{\text{down}}) \cdot S_0 \cdot d - \lambda \cdot S_0 = 0 \\ &= \mathbb{E}^P[u'(V_1) \cdot S_1] = \lambda S_0 \end{aligned}$$

Isolating S_0 :

$$\longrightarrow S_0 = \frac{1}{\lambda} \cdot \mathbb{E}^P[u'(V_1) \cdot S_1] \tag{I}$$

$$\begin{aligned} \mathcal{L}_\beta &= p \cdot u'(X_{\text{up}}) \cdot B_1 + (1 - p) \cdot u'(X_{\text{down}}) \cdot B_1 - \lambda \cdot B_0 = 0 \\ &= \mathbb{E}^P[u'(V_1) \cdot B_1] = \lambda B_0 \end{aligned}$$

Isolating λ :

$$\longrightarrow \lambda = \frac{\mathbb{E}^P[u'(V_1) \cdot B_1]}{B_0} \quad (\text{II})$$

We normalize the bond price to $B_0 = 1$ (one unit invested today) and $B_1 = (1 + r)$ (grows at the risk-free rate):

$$\frac{1}{\lambda} = \frac{1}{(1 + r) \cdot \mathbb{E}^P[u'(V_1)]}$$

Substituting this expression for $\frac{1}{\lambda}$ into (I):

$$S_0 = \frac{1}{1 + r} \cdot \frac{\mathbb{E}^P[u'(V_1) \cdot S_1]}{\mathbb{E}^P[u'(V_1)]} \quad (\text{III})$$

We define the variable Z , which represents the marginal utility of V_1 divided by the expected value of the marginal utility of V_1 :

$$Z = \frac{u'(V_1)}{\mathbb{E}^P[u'(V_1)]}$$

Since V_1 takes two values in the binomial model, Z takes two values accordingly:

$$Z_{\text{up}} = \frac{u'(X_{\text{up}})}{p \cdot u'(X_{\text{up}}) + (1 - p) \cdot u'(X_{\text{down}})} \quad Z_{\text{down}} = \frac{u'(X_{\text{down}})}{p \cdot u'(X_{\text{up}}) + (1 - p) \cdot u'(X_{\text{down}})}$$

Z satisfies two properties:

1. Since utility is increasing, $u'(V_1) > 0$, so $Z > 0$ in both states.
2. The expected value of Z is equal to 1 by construction:

$$\mathbb{E}^P[Z] = \mathbb{E}^P\left[\frac{u'(V_1)}{\mathbb{E}^P[u'(V_1)]}\right] = \frac{\mathbb{E}^P[u'(V_1)]}{\mathbb{E}^P[u'(V_1)]} = 1$$

Using the two properties of Z , we can rewrite equation (III) as:

$$S_0 = \frac{1}{1 + r} \cdot \mathbb{E}^P[Z \cdot S_1]$$

Expanding $\mathbb{E}^P[Z \cdot S_1]$ explicitly in the binomial model, where $S_1 = S_0 u$ in the up state and $S_1 = S_0 d$ in the down state:

$$\begin{aligned} \mathbb{E}^P[Z \cdot S_1] &= p \cdot Z_{\text{up}} \cdot S_0 \cdot u + (1 - p) \cdot Z_{\text{down}} \cdot S_0 \cdot d \\ \longrightarrow S_0 &= \frac{1}{1 + r} \cdot (p \cdot Z_{\text{up}} \cdot S_0 \cdot u + (1 - p) \cdot Z_{\text{down}} \cdot S_0 \cdot d) \end{aligned}$$

By dividing both sides by S_0 , we get:

$$1 + r = p \cdot Z_{\text{up}} \cdot u + (1 - p) \cdot Z_{\text{down}} \cdot d \quad (\text{IV})$$

We define q and $(1 - q)$ as the real-world probability reweighted by how much the investor values a dollar in that state, which makes the investor indifferent between the stock and the risk-free bond:

$$q = p \cdot Z_{\text{up}} = p \cdot \frac{u'(X_{\text{up}})}{E[u'(V_1)]} \quad \text{and} \quad 1 - q = (1 - p) \cdot Z_{\text{down}} = (1 - p) \cdot \frac{u'(X_{\text{down}})}{E[u'(V_1)]}$$

This is valid probability measure because both terms are strictly positive (since $Z > 0$, $p \in (0, 1)$), and they sum to 1:

$$q + (1 - q) = p \cdot Z_{\text{up}} + (1 - p) \cdot Z_{\text{down}} = \mathbb{E}^P[Z] = 1$$

Substituting into equation (IV) and solving for q :

$$1 + r = q \cdot u + (1 - q) \cdot d$$

$$1 + r = q(u - d) + d$$

$$\boxed{q = \frac{(1 + r) - d}{u - d}}$$

This is the **risk-neutral probability**: the unique value of q consistent with no-arbitrage, independent of any investor's preferences.